Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | Portfolio | Virginia, USA

PROFESSIONAL SUMMARY

Software engineer with 8 years of committed, production-focused delivery at a single employer - shipping 20+ full-stack web application modules across 120+ Agile sprints in JavaScript, TypeScript, Python, and PHP for a real estate SaaS platform serving 100K+ monthly active users. Now building AI-powered applications: **AgentProbe** (React frontend + 15-endpoint FastAPI backend, CI-integrated LLM evaluation framework) and **Leasel** (8-agent AI workflow automation platform with LangGraph, intelligent agent orchestration, and HITL escalation). Strong on RESTful API integration, unit and integration testing, cross-functional Agile delivery, and mentoring junior engineers. PropTech/real estate domain expertise from Entrata directly mirrors CoStar's data environment. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). [AgentProbe: github.com/venkatalakshmiparimala/AgentProbe]

CORE COMPETENCIES

React · TypeScript · JavaScript · HTML/CSS | AI Agents · LLM Integration · Workflow Automation | RESTful APIs · FastAPI · Full-Stack Delivery | 8-Year Single-Employer Track Record | Unit & Integration Testing · CI/CD | 120+ Agile Sprints · Mentoring · Code Reviews | PropTech / Real Estate Domain · AWS

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern (Unpaid)

- Built an intelligent agent-based workflow (LangGraph stateful graph: parse -> plan -> retrieve -> validate -> conditional routing) integrating LLM reasoning with deterministic guardrails - translating natural-language requests into structured outputs with prompt engineering and agent orchestration.
- Implemented a RAG layer over FAA Part 107 corpus for cited, grounded answers; engineered validation guardrails enforcing altitude limits, battery budgets, and geometric no-fly-zone checks before export.

George Mason University - College of Engineering

Aug 2025 - May 2026 · Part-time

Graduate Teaching Assistant

- Delivered technical workshops and one-on-one debugging sessions to 600+ students across multiple sections covering Python programming, data structures, OOP, and database design - communicated complex technical concepts clearly to non-technical audiences, adapting explanations in real time.

George Mason University - Research

Aug 2024 - Aug 2025

Research Associate

- Built multi-source prediction models (Python, R) across 190+ countries - $R^2 = 0.977$; applied Isolation Forest for automated anomaly detection across 3 data pipelines; designed Tableau and ggplot2 dashboards improving research comprehension 40%.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 - Aug 2024 · Remote

Senior Software Engineer

- **Shipped 20+ full-stack production modules** (Python, PHP, PostgreSQL, JavaScript, TypeScript, RabbitMQ) across 120+ Agile sprints on a real estate platform serving 100K+ monthly active users - led microservices redesign with Docker and CI/CD pipelines delivering 50% throughput improvement and 30% deployment time reduction.
- **Built and owned production RESTful APIs and backend services** for eligibility, rent, and compliance workflows - integrated across product, compliance, and operations teams; 80% reduction in manual workload through automation pipelines.
- **Owned a mission-critical data pipeline for 7 years** - federal source rows transformed into ~180,000 domain records powering eligibility and tax-credit calculations for 100K+ housing units; every release included unit tests, baseline validation, reconciliation, and rollback preparation. Zero compliance violations.
- **Optimized 200+ SQL queries (30% performance gain)** through execution plan analysis, indexing, partitioning, and stored procedure refactoring across scalable multi-tenant production schemas.
- **Led AWS S3 document migration** for 100K+ user records - 100% PHI compliance with Libsodium-encrypted pipelines; zero security incidents.
- Mentored 5+ junior engineers; conducted 300+ peer code reviews in Agile ceremonies; built internal debugging tools reducing debugging time 40% and keeping open bugs below 10 for 6+ consecutive months.

Entrata India (Db Xento Systems)

Jan 2017 - Sep 2018 · Pune

Associate / Trainee Software Engineer

- Built property management modules (PHP, Python, PostgreSQL) for compliance workflows across Agile sprints; led AWS S3 cloud migration; redesigned document storage database architecture for 180K+ records.

PROJECTS

AgentProbe: AI Agent Evaluation Framework

Open Source

2025 | Python, React, TypeScript, FastAPI, LangGraph, ChromaDB, CI/CD (GitHub Actions)

github.com/venkatalakshmiparimala/AgentProbe

Built a full-stack AI evaluation platform: React frontend for agent monitoring and test visualization, 15-endpoint FastAPI backend, 5-metric eval engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate blocking quality regressions before merge, and multi-provider LLM routing. Unit and integration tests via automated CI pipeline; plug-and-play REST API - any agent connects without code changes. Validated LeaseI across 1,240+ test cases: pass@1 81.7%, pass@3 88.3%.

LeaseI: AI Workflow Automation Platform

Intelligent Agents

2026 | Python, LangGraph (8-agent), React, FastAPI, RAG, HITL, OCR, SQLite

github.com/venkatalakshmiparimala/Lease_complaince_rag

Built an enterprise AI workflow automation platform with 8 intelligent agents (LangGraph orchestration: Triage -> Retrieval -> Configuration -> Reasoning -> Validation -> Human Escalation), React-powered engineer console, and HITL escalation when agent confidence is too low. Integrated AI agents into internal compliance workflows - LLM reasoning feeds into deterministic decision logic. Results: 93.6% precision, 87.9% F1 on 1,240 compliance documents; 100% citation correctness; 0% unsupported-claim rate.

AI on FHIR: NLP-Powered Medical Records Query System

2024 | Flask, React, LLMs, FHIR API, Docker, spaCy, BERT, Embeddings, JavaScript

github.com/venkatalakshmiparimala/AI_on_FHIR

Built a Flask + React full-stack AI application translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings - reduced manual clinical retrieval time ~60%; enforced HIPAA-compliant RBAC; containerized with Docker.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering - Jawaharlal Nehru Technological University, Anantapur — 2017

SKILLS

Frontend: React.js, TypeScript, JavaScript, HTML, CSS, Responsive Design, REST API Integration, jQuery, AJAX

AI & Agents: LLM Integration, Prompt Engineering, Agent-Based Architectures, Workflow Automation, LangGraph, Intelligent Agents, HITL, RAG Pipelines, Embeddings, OpenAI API, Anthropic API, FastAPI

Backend & Data: Python, PHP, PostgreSQL, MySQL, Oracle, SQL, PL/SQL, RabbitMQ, Microservices, RESTful APIs, Docker, AWS (S3, EC2, RDS, Glue)

Engineering Practices: CI/CD (GitHub Actions), Unit Testing, Integration Testing, Code Reviews, Agile/Scrum, Git, Linux, Performance Engineering, Scalable Systems